A Monero data model and API specification

with implementations in Python, Java, TypeScript, and C++
using monero v0.18.5.3 'Fluorine Fermi'

Last updated November 1, 2026



Python implementation: https://github.com/libmonero/monero-python

Java implementation: https://github.com/woodser/monero-java
TypeScript implementation: https://github.com/woodser/monero-ts
C++ implementation: https://github.com/woodser/monero-cpp
C# implementation: https://github.com/libmonero/monero-csharp
Kotlin implementation: https://github.com/libmonero/monero-kotlin

Document source: https://github.com/libmonero/monero-python/blob/main/docs/monero-spec.drawio

Document created and revisable using: https://www.draw.io/

Monero Daemon Interface 1/3

```
get_version() -> MoneroVersion

is_trusted() -> bool

get_height() -> int

get_block_hash(height: int) -> str

get_block_template(wallet_address: str, reserve_size: Optional[int]=None) -> MoneroBlockTemplate

get_miner_data() -> MoneroMinerData

get_last_block_header() -> MoneroBlockHeader

get_block_header_by_hash(hash: str) -> MoneroBlockHeader

get_block_header_by_height(height: int) -> MoneroBlockHeader

get_block_headers_by_range(start_height: int, end_height: int) -> list[MoneroBlockHeader]

get_block_by_hash(hash: str) -> MoneroBlock

get_blocks_by_hash(block_hashes: list[str], start_height: int, prune: bool, max_block_count: int=0) -> MoneroGetBlocksByHashResult

get_block_by_height(height: int) -> MoneroBlock

get_blocks_by_height(heights: list[int]) -> list[MoneroBlock]

get_blocks_by_range(start_height: Optional[int], end_height: Optional[int]) -> list[MoneroBlock]

get_blocks_by_range_chunked(start_height: Optional[int], end_height: Optional[int], max_chunk_size: Optional[int]=None)
    -> list[MoneroBlock]

get_block_hashes(block_hashes: list[str]) -> MoneroGetBlockHashesResult

get_tx(tx_hash: str, prune: bool=False) -> Optional[MoneroTx]

get_txs(tx_hashes: list[str], prune: bool=False) -> list[MoneroTx]

get_tx_hex(tx_hash: str, prune: bool=False) -> Optional[str]

get_tx_hexes(tx_hashes: list[str], prune: bool=False) -> list[str]

get_miner_tx_sum(height: int, num_blocks: int) -> MoneroMinerTxSum

get_fee_estimate(grace_blocks: int=0) -> MoneroFeeEstimate

submit_tx_hex(tx_hex: str, do_not_relay: bool=False) -> MoneroSubmitTxResult

relay_tx_by_hash(tx_hash: str) -> None

relay_txs_by_hash(tx_hashes: list[str]) -> None

get_tx_pool() -> list[MoneroTx]

get_tx_pool_hashes() -> list[str]
```

Monero Daemon Interface 2/3

get_tx_pool_backlog() -> list[MoneroTxBacklogEntry]

get_tx_pool_stats() -> MoneroTxPoolStats

flush_tx_pool() -> None

flush_tx_pool(hash: str) -> None

flush_tx_pool(hashes: list[str]) -> None

flush_cache(bad_blocks: bool=False) -> None

get_key_image_spent_status(key_image: str) -> MoneroKeyImageSpentStatus

get_key_image_spent_statuses(key_images: list[str]) -> list[MoneroKeyImageSpentStatus]

get_outputs(outputs: list[MoneroOutput]) -> list[MoneroOutput]

get_output_indices(tx_hash: str) -> list[int]

get_info() -> MoneroDaemonInfo

get_sync_info() -> MoneroDaemonSyncInfo

get_hard_fork_info() -> MoneroHardForkInfo

get_alt_chains() -> list[MoneroAltChain]

get_alt_block_hashes() -> list[str]

get_network_stats() -> MoneroDaemonNetworkStats

get_download_limit() -> int

set_download_limit(limit: int) -> int

reset_download_limit() -> int

get_upload_limit() -> int

set_upload_limit(limit: int) -> int

reset_upload_limit() -> int

get_peers() -> list[MoneroPeer]

get_known_peers() -> list[MoneroPeer]

get_public_peers(include_offline: bool=False) -> list[MoneroPeer]

set_outgoing_peer_limit(limit: int) -> None

set_incoming_peer_limit(limit: int) -> None

get_peer_bans() -> list[MoneroBan]

get_peer_ban(address: str) -> MoneroBan

Monero Daemon Interface 3/3

```
set_peer_ban(ban: MoneroBan) -> None

set_peer_bans(bans: list[MoneroBan]) -> None

get_output_distribution(amounts: list[int], is_cumulative: Optional[bool]=None, start_height: Optional[int]=None,
    end_height: Optional[int]=None) -> list[MoneroOutputDistributionEntry]

get_output_histogram(amounts: list[int], min_count: Optional[int], max_count: Optional[int], is_unlocked: Optional[bool],
    recent_cutoff: Optional[int]) -> list[MoneroOutputHistogramEntry]

start_mining(address: str, num_threads: int, is_background: bool, ignore_battery: bool) -> None

stop_mining() -> None

get_mining_status() -> MoneroMiningStatus

submit_block(block_blob: str) -> None

submit_blocks(block_blobs: list[str]) -> None

generate_blocks(wallet_address: str, num_blocks: int, prev_block_hash: Optional[str]=None, starting_nonce: Optional[int]=None)
    -> MoneroGenerateBlocksResult

check_for_update() -> MoneroDaemonUpdateCheckResult

download_update(path: str='') -> MoneroDaemonUpdateDownloadResult

set_bootstrap_daemon(address: str, username: str='', password: str='', proxy: str='') -> None

add_listener(listener: MoneroDaemonListener) -> None

get_listeners() -> list[MoneroDaemonListener]

remove_listener(listener: MoneroDaemonListener) -> None

remove_listeners() -> None

wait_for_next_block_header() -> MoneroBlockHeader

stop() -> None

prune_blockchain(check: bool) -> MoneroPruneResult

pop_blocks(num_blocks: int) -> int

save_blockchain() -> None

add_auxiliary_pow(block_template_blob: str, aux_pow: list[MoneroAuxiliaryPow]) -> MoneroAddAuxiliaryPowResult

calculate_pow(major_version: int, height: int, block_blob: str, seed_hash: str) -> str

remove_bootstrap_daemon() -> None

set_log_level(level: int) -> None

set_log_categories(categories: str='') -> str

set_log_hash_rate(is_visible: bool) -> None
```

Monero Daemon Types 1/3



Monero Daemon Types 1/3 (continued)

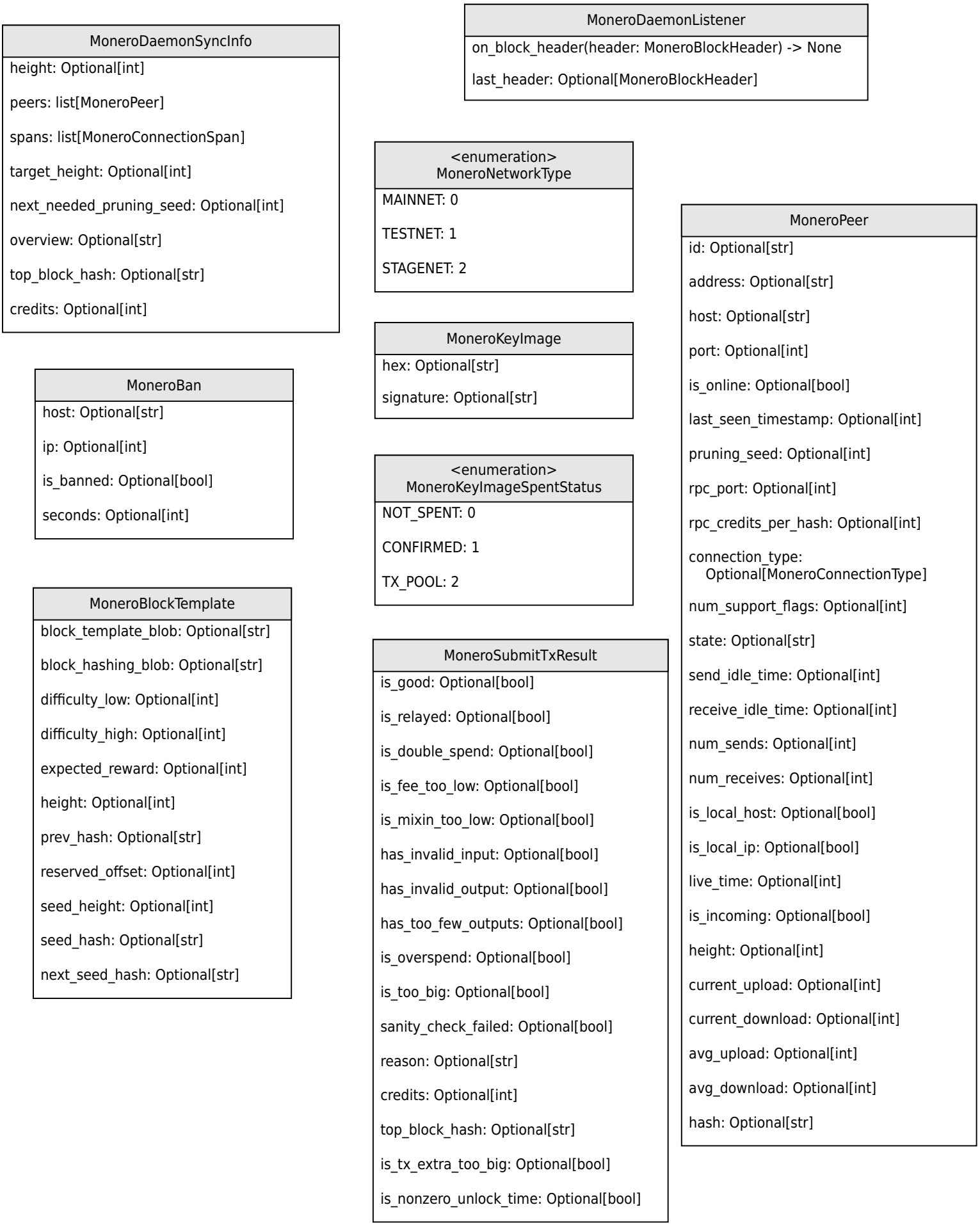
MoneroOutput
tx: MoneroTx key_image: Optional[MoneroKeyImage] amount: Optional[int] index: Optional[int] ring_output_indices: list[int] stealth_public_key: Optional[str] mask: Optional[str]

MoneroGetBlockHashesResult
hashes: list[str] start_height: Optional[int] current_height: Optional[int]

MoneroGetBlocksByHashResult
blocks: list[MoneroBlock] current_height: Optional[int]

MoneroRpcPaymentInfo
credits: Optional[int] top_block_hash: Optional[str]

Monero Daemon Types 2/3



Monero Daemon Types 2/3 (continued)

MoneroConnectionSpan

- connection_id: Optional[str]
- num_blocks: Optional[int]
- remote_address: Optional[str]
- rate: Optional[int]
- speed: Optional[int]
- size: Optional[int]
- start_height: Optional[int]

MoneroDaemonNetworkStats
extends MoneroRpcPaymentInfo

- start_time: Optional[int]
- total_packets_in: Optional[int]
- total_bytes_in: Optional[int]
- total_packets_out: Optional[int]
- total_bytes_out: Optional[int]

MoneroRpcConnection

- priority: int
- proxy_uri: Optional[str]
- zmq_uri: Optional[str]
- timeout_ms: int
- ssl_verify: bool
- uri: Optional[str]
- username: Optional[str]
- password: Optional[str]
- response_time: Optional[int]
- compare(p1: int, p2: int) -> bool
- check_connection(
  timeout_ms: Optional[int]=None)
  -> bool
- get_attribute(key: str) -> str
- is_authenticated() -> Optional[bool]
- is_connected() -> Optional[bool]
- is_i2p() -> bool
- is_onion() -> bool
- is_online() -> Optional[bool]
- send_json_request(method: str,
  parameters: Optional[object]=None,
  timeout_ms: Optional[int]=None)
  -> Optional[object]
- send_path_request(method: str,
  parameters: Optional[object]=None,
  timeout_ms: Optional[int]=None)
  -> Optional[object]
- send_binary_request(method: str,
  parameters: Optional[object]=None,
  timeout_ms: Optional[int]=None)
  -> Optional[bytes]
- set_attribute(key: str, value: str) -> None
- set_credentials(username: str,
  password: str) -> None

MoneroAddAuxiliaryPowResult
extends MoneroRpcPaymentInfo

- block_template_blob: Optional[str]
- block_hashing_blob: Optional[str]
- merkle_root: Optional[str]
- merkle_tree_depth: Optional[int]
- aux_pow: list[MoneroAuxiliaryPow]

MoneroAuxiliaryPow

- id: Optional[str]
- hash: Optional[str]

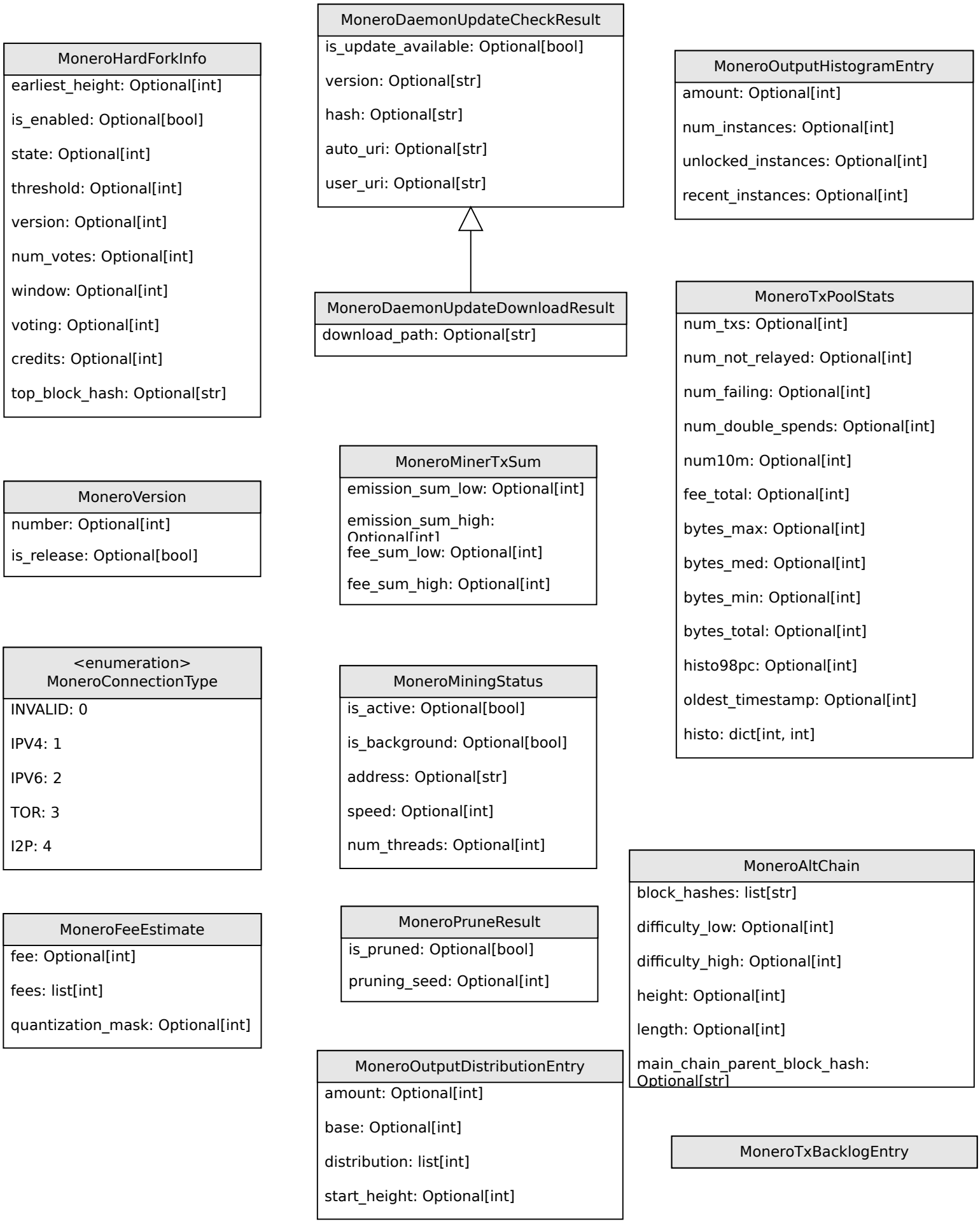
MoneroGenerateBlocksResult

- block_hashes: list[str]
- height: Optional[int]

MoneroMinerData
extends MoneroRpcPaymentInfo

- major_version: Optional[int]
- height: Optional[int]
- prev_hash: Optional[str]
- seed_hash: Optional[str]
- difficulty: Optional[str]
- median_weight: Optional[int]
- already_generated_coins: Optional[int]
- tx_pool_backlog: list[MoneroTx]

Monero Daemon Types 3/3



Monero Wallet Interface 1/4

DEFAULT_LANGUAGE: str

add_listener(listener: MoneroWalletListener) -> None

remove_listener(listener: MoneroWalletListener) -> None

get_listeners() -> list[MoneroWalletListener]

is_view_only() -> bool

set_daemon_connection(connection: Optional[MoneroRpcConnection], is_trusted: Optional[bool]=None) -> None

set_daemon_connection(uri: str, username: str='', password: str='', proxy_uri: str='', is_trusted: Optional[bool]=None) -> None

get_daemon_connection() -> Optional[MoneroRpcConnection]

is_connected_to_daemon() -> bool

is_daemon_synced() -> bool

is_daemon_trusted() -> bool

is_synced() -> bool

get_version() -> MoneroVersion

get_network_type() -> MoneroNetworkType

get_path() -> str

get_seed() -> str

get_seed_language() -> str

get_private_spend_key() -> str

get_private_view_key() -> str

get_public_spend_key() -> str

get_public_view_key() -> str

get_primary_address() -> str

get_address(account_idx: int, subaddress_idx: int) -> str

get_address_index(address: str) -> MoneroSubaddress

get_daemon_height() -> int

get_daemon_max_peer_height() -> int

get_height() -> int

get_height_by_date(year: int, month: int, day: int) -> int

get_restore_height() -> int

set_restore_height(restore_height: int) -> None

sync() -> MoneroSyncResult

sync(listener: MoneroWalletListener) -> MoneroSyncResult

sync(start_height: int) -> MoneroSyncResult

sync(start_height: int, listener: MoneroWalletListener) -> MoneroSyncResult

start_syncing(sync_period_in_ms: int=10000) -> None

Monero Wallet Interface 2/4

```
stop_syncing() -> None
wait_for_next_block() -> int
scan_txs(tx_hashes: list[str]) -> None
rescan_spent() -> None
rescan_blockchain() -> None
get_balance() -> int
get_balance(account_idx: int) -> int
get_balance(account_idx: int, subaddress_idx: int) -> int
get_unlocked_balance() -> int
get_unlocked_balance(account_idx: int) -> int
get_unlocked_balance(account_idx: int, subaddress_idx: int) -> int
get_accounts() -> list[MoneroAccount]
get_accounts(include_subaddresses: bool) -> list[MoneroAccount]
get_accounts(tag: str) -> list[MoneroAccount]
get_accounts(include_subaddresses: bool, tag: str) -> list[MoneroAccount]
get_account(account_idx: int) -> MoneroAccount
get_account(account_idx: int, include_subaddresses: bool) -> MoneroAccount
create_account(label: str='') -> MoneroAccount
get_subaddresses(account_idx: int) -> list[MoneroSubaddress]
get_subaddresses(account_idx: int, subaddress_indices: list[int]) -> list[MoneroSubaddress]
get_subaddress(account_idx: int, subaddress_idx: int) -> MoneroSubaddress
create_subaddress(account_idx: int, label: str='') -> MoneroSubaddress
set_account_label(account_idx: int, label: str) -> None
set_subaddress_label(account_idx: int, subaddress_idx: int, label: str='') -> None
get_tx(tx_hash: str) -> Optional[MoneroTxWallet]
get_txs() -> list[MoneroTxWallet]
get_txs(tx_hashes: list[str]) -> list[MoneroTxWallet]
get_txs(query: MoneroTxQuery) -> list[MoneroTxWallet]
get_transfers() -> list[MoneroTransfer]
get_transfers(query: MoneroTransferQuery) -> list[MoneroTransfer]
get_transfers(account_idx: int) -> list[MoneroTransfer]
get_transfers(account_idx: int, subaddress_idx: int) -> list[MoneroTransfer]
get_outputs() -> list[MoneroOutputWallet]
get_outputs(query: MoneroOutputQuery) -> list[MoneroOutputWallet]
export_outputs(all: bool=False) -> str
```

Monero Wallet Interface 3/4

import_outputs(outputs_hex: str) -> int

export_key_images(all: bool=False) -> MoneroKeyImageExportResult

import_key_images(key_images: list[MoneroKeyImage], offset: int=0) -> MoneroKeyImageImportResult

freeze_output(key_image: str) -> None

thaw_output(key_image: str) -> None

is_output_frozen(key_image: str) -> bool

get_new_key_images_from_last_import() -> MoneroKeyImageExportResult

get_default_fee_priority() -> MoneroTxPriority

create_tx(config: MoneroTxConfig) -> MoneroTxWallet

create_txs(config: MoneroTxConfig) -> list[MoneroTxWallet]

sweep_output(config: MoneroTxConfig) -> MoneroTxWallet

sweep_unlocked(config: MoneroTxConfig) -> list[MoneroTxWallet]

sweep_dust(relay: bool=False) -> list[MoneroTxWallet]

relay_tx(tx_metadata: str) -> str

relay_tx(tx: MoneroTxWallet) -> str

relay_txs(txs: list[MoneroTxWallet]) -> list[str]

relay_txs(tx_metadatas: list[str]) -> list[str]

describe_tx_set(tx_set: MoneroTxSet) -> MoneroTxSet

describe_unsigned_tx_set(unsigned_tx_hex: str) -> MoneroTxSet

describe_multisig_tx_set(multisig_tx_hex: str) -> MoneroTxSet

sign_txs(unsigned_tx_hex: str) -> MoneroTxSet

submit_txs(signed_tx_hex: str) -> list[str]

sign_message(msg: str, signature_type: MoneroMessageSignatureType, account_idx: int=0, subaddress_idx: int=0) -> str

verify_message(msg: str, address: str, signature: str) -> MoneroMessageSignatureResult

get_tx_key(tx_hash: str) -> str

check_tx_key(tx_hash: str, tx_key: str, address: str) -> MoneroCheckTx

get_tx_proof(tx_hash: str, address: str, message: str='') -> str

check_tx_proof(tx_hash: str, address: str, message: str, signature: str) -> MoneroCheckTx

get_spend_proof(tx_hash: str, message: str='') -> str

check_spend_proof(tx_hash: str, message: str, signature: str) -> bool

get_reserve_proof_wallet(message: str) -> str

get_reserve_proof_account(account_idx: int, amount: int, message: str) -> str

check_reserve_proof(address: str, message: str, signature: str) -> MoneroCheckReserve

set_tx_notes(tx_hashes: list[str], notes: list[str]) -> None

set_tx_note(tx_hash: str, note: str) -> None

Monero Wallet Interface 4/4

get_tx_notes(tx_hashes: list[str]) -> list[str]

get_tx_note(tx_hash: str) -> str

get_address_book_entries() -> list[MoneroAddressBookEntry]

get_address_book_entries(indices: list[int]) -> list[MoneroAddressBookEntry]

add_address_book_entry(address: str, description: str) -> int

edit_address_book_entry(index: int, set_address: bool, address: str, set_description: bool, description: str) -> None

delete_address_book_entry(index: int) -> None

tag_accounts(tag: str, account_indices: list[int]) -> None

untag_accounts(account_indices: list[int]) -> None

get_account_tags() -> list[MoneroAccountTag]

set_account_tag_label(tag: str, label: str) -> None

get_payment_uri(config: MoneroTxConfig) -> str

parse_payment_uri(uri: str) -> MoneroTxConfig

get_attribute(key: str) -> str

set_attribute(key: str, val: str) -> None

start_mining(num_threads: Optional[int]=None, background_mining: Optional[bool]=None, ignore_battery: Optional[bool]=None) -> None

stop_mining() -> None

is_multisig_import_needed() -> bool

is_multisig() -> bool

get_multisig_info() -> MoneroMultisigInfo

prepare_multisig() -> str

make_multisig(multisig_hexes: list[str], threshold: int, password: str) -> str

exchange_multisig_keys(multisig_hexes: list[str], password: str) -> MoneroMultisigInitResult

export_multisig_hex() -> str

import_multisig_hex(multisig_hexes: list[str], refresh_after_import: bool=True) -> int

sign_multisig_tx_hex(multisig_tx_hex: str) -> MoneroMultisigSignResult

submit_multisig_tx_hex(signed_multisig_tx_hex: str) -> list[str]

move_to(path: str, password: str) -> None

change_password(old_password: str, new_password: str) -> None

save() -> None

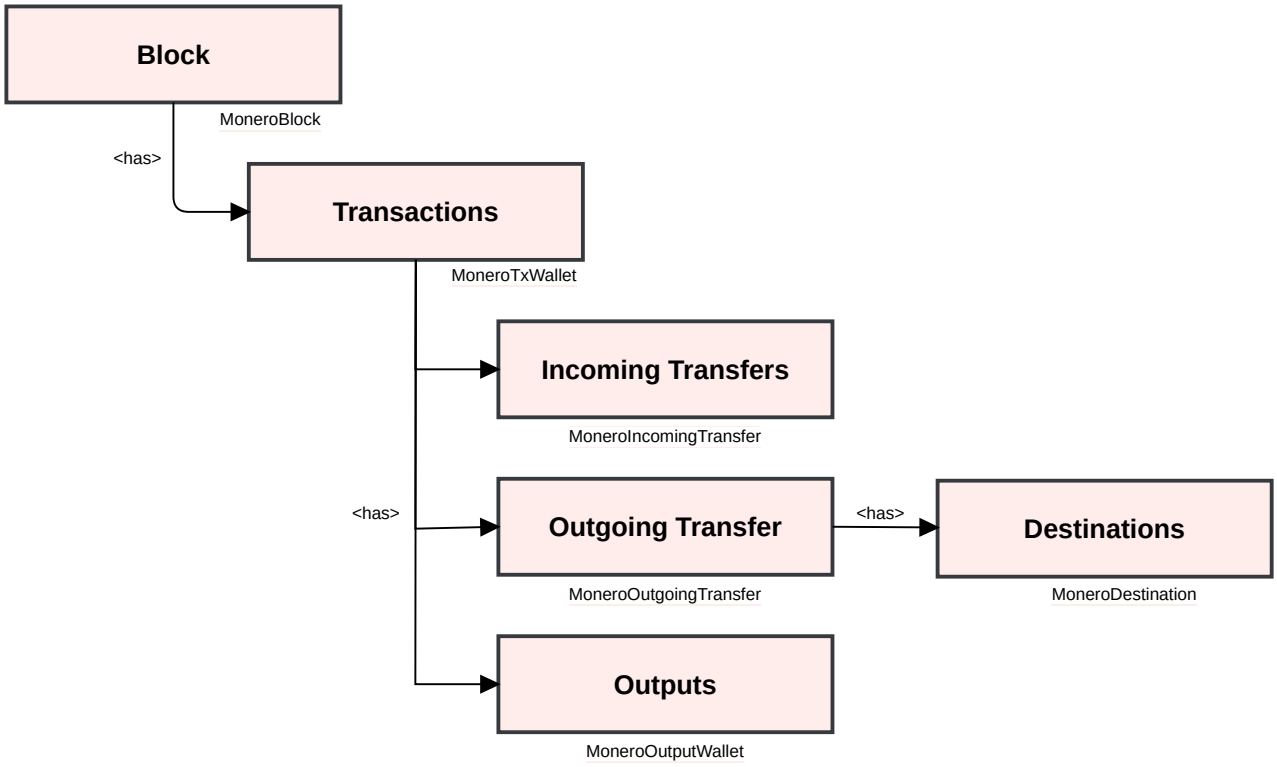
close(save: bool=False) -> None

is_closed() -> bool

get_integrated_address(standard_address: str='', payment_id: str='') -> MoneroIntegratedAddress (deprecated)

decode_integrated_address(integrated_address: str) -> MoneroIntegratedAddress (deprecated)

Monero Wallet Types 1/4 - Transactions, Transfers, and Outputs

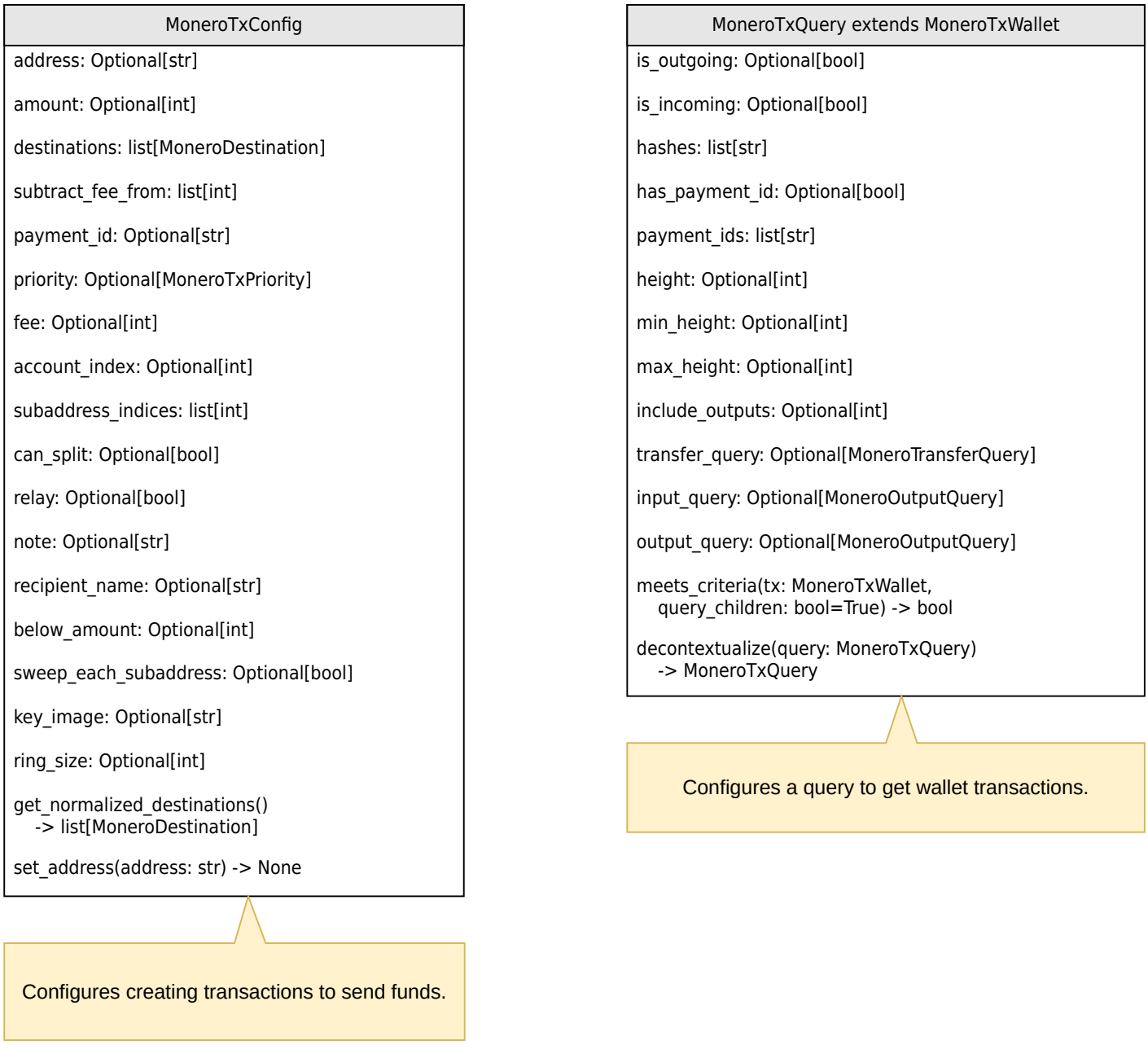


Transactions are part of a block and have incoming transfers and/or an outgoing transfer.

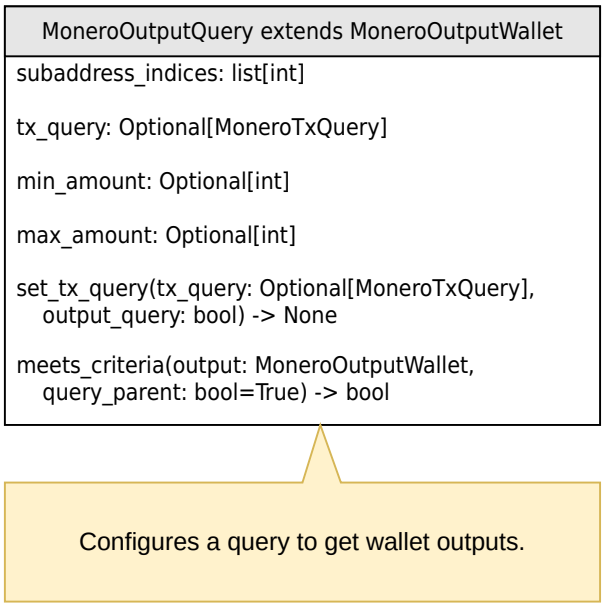
The outgoing transfer may have destination amounts and addresses which are saved locally with the wallet and are not recoverable from the blockchain.

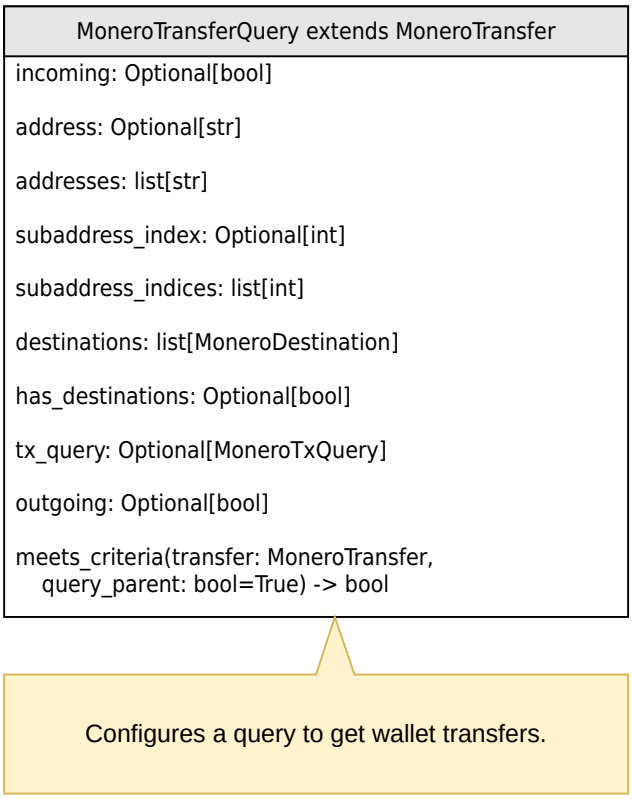
Transactions may also have outputs received by the wallet when requested.

Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs

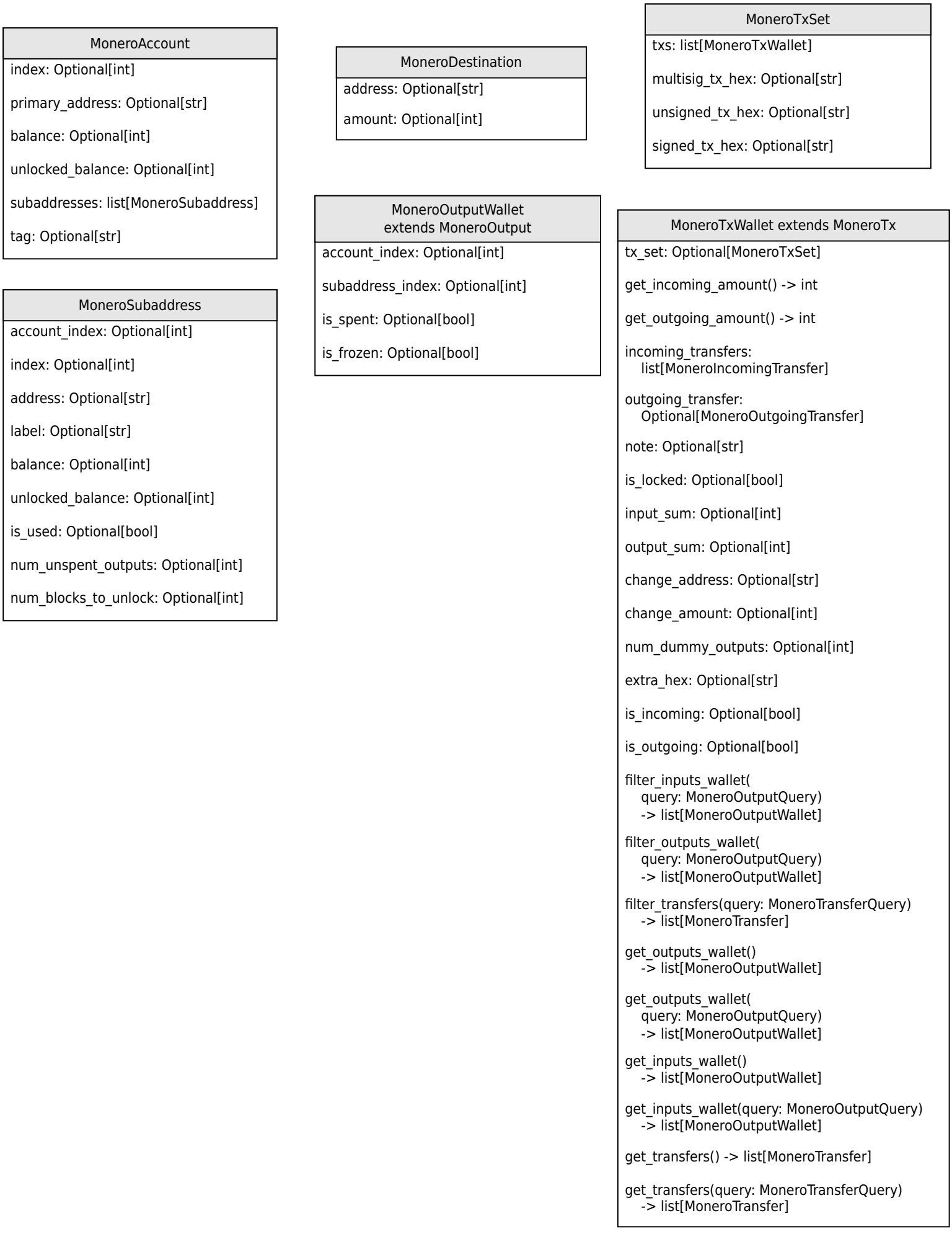


Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs (continued)

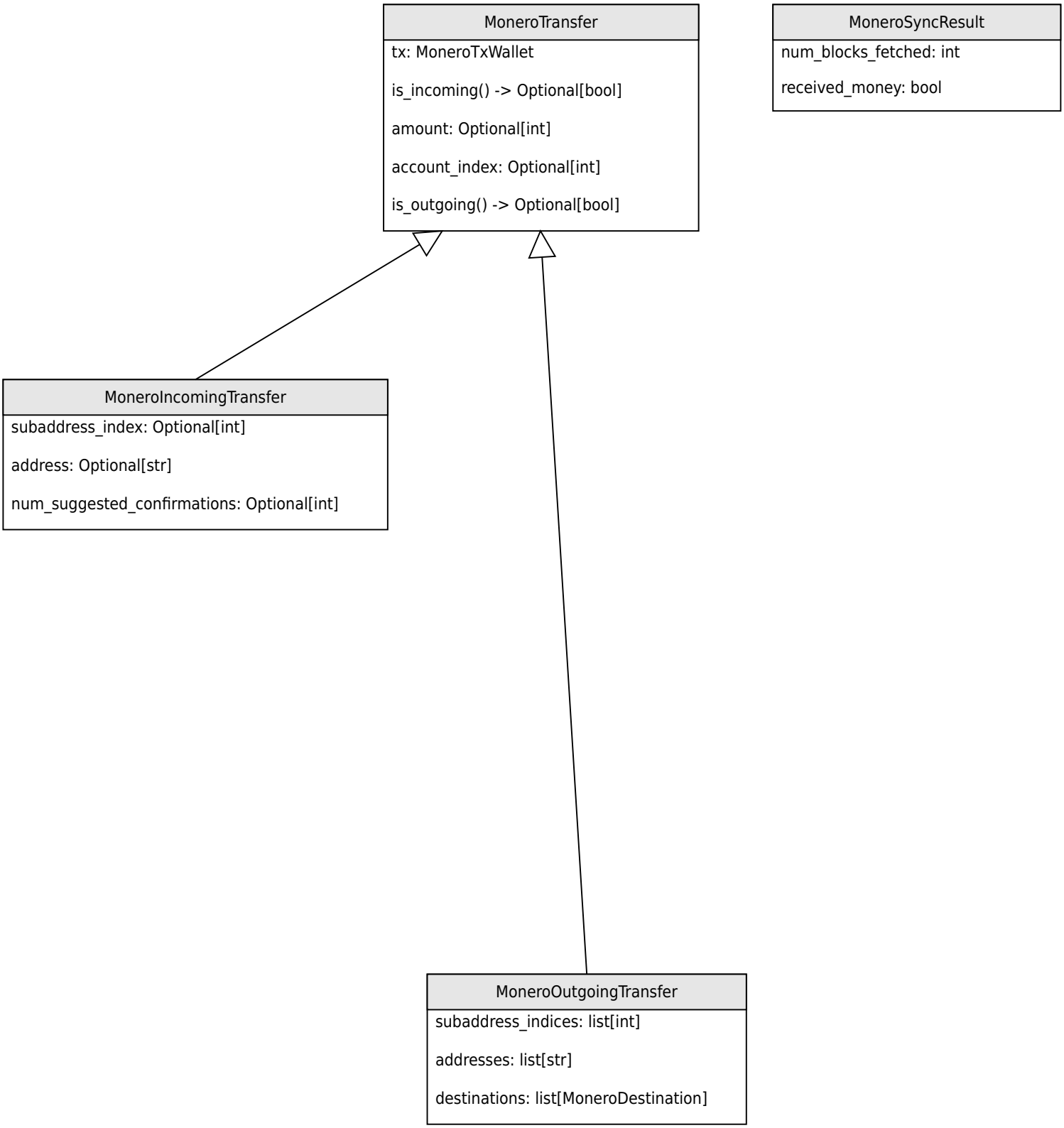




Monero Wallet Types 3/4



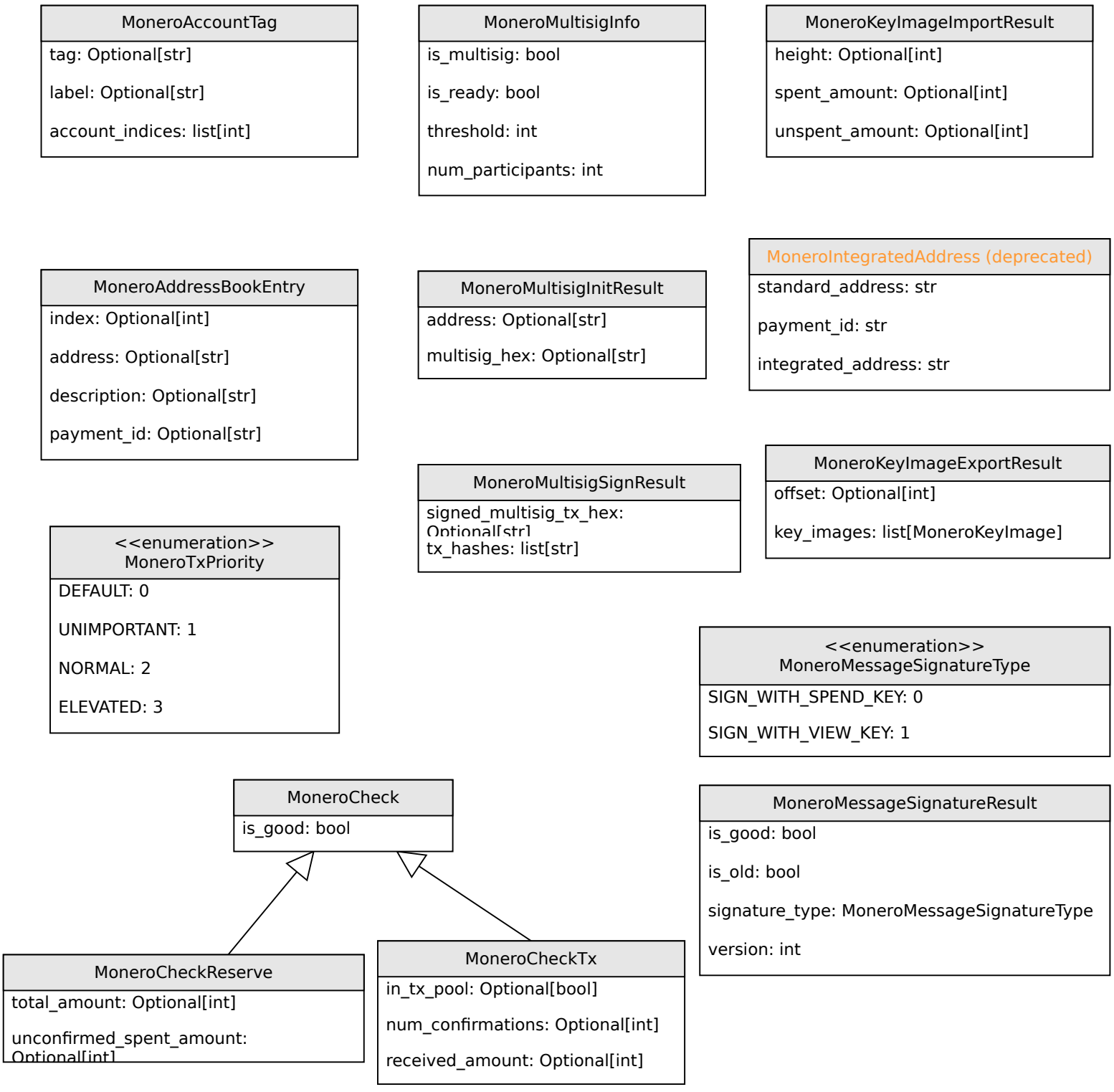
Monero Wallet Types 3/4 (continued)



Monero Wallet Types 3/4 (continued)

MoneroWalletListener
on_sync_progress(height: int, start_height: int, end_height: int, percent_done: float, message: str) -> None
on_new_block(height: int) -> None
on_output_received(output: MoneroOutputWallet) -> None
on_output_spent(output: MoneroOutputWallet) -> None
on_balances_changed(new_balance: int, new_unlocked_balance: int) -> None

Monero Wallet Types 4/4



woodser API 1/1



84rXAu3QhsfbQ2vbxyoEptbRLZ2gtyL8k2qCeJwM5AU4FcxVDEbLRuxMZD5pXD1TR295Nodn7y9mkjoMemHrnMqa9ZnJTza

